

The Heston Stochastic Volatility Model

From volatility smiles to equity skew

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September 2026

Why stochastic volatility?

Black–Scholes assumes that volatility is constant.

For a fixed maturity, this implies that all option prices should produce the same implied volatility.

But equity-option markets look different.

- ▶ implied volatility changes across strikes,
- ▶ lower strikes often carry higher implied volatility,
- ▶ the resulting curve displays a smile or, in equities, a pronounced downward skew.

This motivates models in which variance itself evolves randomly over time.

From constant volatility to Heston

The story begins with a simple progression.

$$\boxed{\text{Constant volatility}} \longrightarrow \boxed{\text{Stochastic variance}}$$

Random variance can generate curvature in the implied-volatility curve. Under an independent-shock stochastic-volatility specification, however, that curvature remains approximately symmetric.

Heston adds a second ingredient:

$$\boxed{\text{Stochastic variance}} + \boxed{\text{Stock–variance correlation}}.$$

With negative stock–variance correlation, the model can generate the asymmetric downward skew characteristic of equity options.

Heston model: stock and variance dynamics

Under the physical measure $\mathbb{P}$, the Heston model assumes

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S,$$

while instantaneous variance follows a CIR process,

$$dv_t = \kappa(\bar{v} - v_t) dt + \zeta \sqrt{v_t} dW_t^v.$$

The Brownian motions satisfy

$$d\langle W^S, W^v \rangle_t = \rho dt, \quad \rho \in [-1, 1].$$

Here v_t is variance; instantaneous volatility is $\sqrt{v_t}$.

Why the CIR variance process?

The drift

$$\kappa(\bar{v} - v_t)$$

pulls variance back toward the long-run level $\bar{v}$.

- ▶ κ : speed of mean reversion,
- ▶ $\bar{v}$: long-run variance level,
- ▶ ζ : volatility of variance, or *vol-of-vol*.

The diffusion term

$$\zeta\sqrt{v_t} dW_t^v$$

shrinks as v_t approaches zero. A standard sufficient condition for strict positivity is the Feller condition

$$2\kappa\bar{v} \geq \zeta^2.$$

Moving to the risk-neutral measure

For option pricing we move from $\mathbb{P}$ to a risk-neutral measure $\mathbb{Q}$.

No-arbitrage fixes the stock drift under $\mathbb{Q}$:

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{S,\mathbb{Q}}.$$

Variance is not directly traded, so its risk-neutral drift is not uniquely fixed by no-arbitrage. We parameterise the volatility-risk premium so that the variance process retains the CIR form under $\mathbb{Q}$.

With the sign convention used here, the variance dynamics under $\mathbb{Q}$ can be written

$$dv_t = \kappa^*(\bar{v}^* - v_t) dt + \zeta\sqrt{v_t} dW_t^{v,\mathbb{Q}}.$$

Risk-neutral variance parameters

Using the sign convention adopted here, preserving the CIR form gives

$$\kappa^* = \kappa + \Lambda\zeta, \quad \bar{v}^* = \frac{\kappa\bar{v}}{\kappa + \Lambda\zeta}.$$

Hence the risk-neutral Heston system is

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^{S,\mathbb{Q}}, \\ dv_t &= \kappa^*(\bar{v}^* - v_t) dt + \zeta\sqrt{v_t} dW_t^{v,\mathbb{Q}}, \end{aligned}$$

with

$$d\langle W^{S,\mathbb{Q}}, W^{v,\mathbb{Q}} \rangle_t = \rho dt.$$

The Heston pricing PDE

Let $C(t, S, v)$ denote the price of a European option with terminal payoff

$$C(T, S, v) = g(S).$$

Under the risk-neutral Heston dynamics,

$$\begin{aligned} C_t + \frac{1}{2}vS^2C_{SS} + \rho\zeta vSC_{Sv} + \frac{1}{2}\zeta^2vC_{vv} \\ + rSC_S - rC + \kappa^*(\bar{v}^* - v)C_v = 0. \end{aligned}$$

The correlation parameter enters explicitly through the mixed derivative

$$\rho\zeta vSC_{Sv}.$$

Step 1: move to log-price coordinates

The coefficients $S^2 C_{SS}$, SC_{Sv} and SC_S depend directly on S .

Set

$$x = \ln S, \quad U(t, x, v) := C(t, e^x, v).$$

Using the chain rule,

$$C_S = \frac{U_x}{S}, \quad C_{SS} = \frac{U_{xx} - U_x}{S^2}, \quad C_{Sv} = \frac{U_{xv}}{S}.$$

The last two identities follow from the product and chain rules; these are the derivative calculations carried out explicitly in the appendix.

The log-price PDE

Substituting the transformed derivatives gives

$$U_t + \frac{1}{2}vU_{xx} + \rho\zeta vU_{xv} + \frac{1}{2}\zeta^2vU_{vv} \\ + \left(r - \frac{v}{2}\right)U_x - rU + \kappa^*(\bar{v}^* - v)U_v = 0.$$

The direct dependence on S has disappeared. The remaining coefficients are constant or affine in v , which is the structure needed for the Fourier–affine solution.

Step 2: decompose the option price

Following Duffie, Pan and Singleton (2000), as presented in Gatheral (2006), introduce

$$C(t, S, v) = SP_1(t, x, v) - Ke^{-r\tau}P_2(t, x, v), \quad \tau = T - t.$$

Since $S = e^x$ and $U(t, x, v) = C(t, e^x, v)$,

$$U(t, x, v) = e^x P_1(t, x, v) - Ke^{-r\tau}P_2(t, x, v).$$

At this stage P_1 and P_2 are unknown functions. The next step is to determine the PDEs they must satisfy.

Terminal condition for P_1 and P_2

For a European call, the Heston representation uses the common terminal condition

$$P_j(x, v, 0) = \chi(x) = \begin{cases} 1, & x > \ln K, \\ 0, & x \leq \ln K, \end{cases} \quad j = 1, 2.$$

Substituting this into the price decomposition at $\tau = 0$ gives

$$SP_1(x, v, 0) - KP_2(x, v, 0) = S\chi(x) - K\chi(x) = (S - K)^+.$$

Thus the boundary condition reproduces the European call payoff exactly.

At expiry the option is either in the money or it is not.

Appendix calculation: differentiating the decomposition

Write $K_\tau := Ke^{-r\tau}$. From

$$U = e^x P_1 - K_\tau P_2,$$

we obtain

$$\begin{aligned}U_t &= e^x P_{1,t} - rK_\tau P_2 - K_\tau P_{2,t}, \\U_x &= e^x P_1 + e^x P_{1,x} - K_\tau P_{2,x}, \\U_{xx} &= e^x P_1 + 2e^x P_{1,x} + e^x P_{1,xx} - K_\tau P_{2,xx}, \\U_v &= e^x P_{1,v} - K_\tau P_{2,v}, \\U_{vv} &= e^x P_{1,vv} - K_\tau P_{2,vv}, \\U_{xv} &= e^x P_{1,v} + e^x P_{1,xv} - K_\tau P_{2,xv}.\end{aligned}$$

These are the product-rule calculations used to separate the e^x and K_τ components of the PDE.

Step 3: the PDEs for P_j

After substitution and collecting the two independent components, both equations can be written as

$$\begin{aligned} P_{j,t} + \frac{1}{2}vP_{j,xx} + \rho\zeta vP_{j,xv} + \frac{1}{2}\zeta^2vP_{j,vv} \\ + (r + u_jv)P_{j,x} + (a - b_jv)P_{j,v} = 0, \end{aligned}$$

where

$$u_1 = \frac{1}{2}, \quad u_2 = -\frac{1}{2}, \quad a = \kappa^*\bar{v}^*, \quad b_1 = \kappa^* - \rho\zeta, \quad b_2 = \kappa^*.$$

Both share the terminal condition $P_j(x, v, 0) = \chi(x)$ once the problem is written in time-to-maturity.

Step 4: turn the terminal problem into an initial problem

Set

$$\tau = T - t.$$

Then $t = T$ corresponds to $\tau = 0$, and

$$P_{j,t} = P_{j,\tau} \frac{\partial \tau}{\partial t} = -P_{j,\tau}.$$

Hence

$$\begin{aligned} -P_{j,\tau} + \frac{1}{2}vP_{j,xx} + \rho\zeta vP_{j,xv} + \frac{1}{2}\zeta^2vP_{j,vv} \\ + (r + u_jv)P_{j,x} + (a - b_jv)P_{j,v} = 0, \end{aligned}$$

with initial condition $P_j(x, v, 0) = \chi(x)$.

Step 5: transform into Fourier space

Define the Fourier transform in the log-price variable x by

$$\hat{P}_j(\phi, v, \tau) = \int_{-\infty}^{\infty} e^{-i\phi x} P_j(x, v, \tau) dx.$$

The standard derivative identities give

$$\mathcal{F}[P_{j,x}] = i\phi \hat{P}_j, \quad \mathcal{F}[P_{j,xx}] = -\phi^2 \hat{P}_j, \quad \mathcal{F}[P_{j,xv}] = i\phi \hat{P}_{j,v}.$$

So every derivative with respect to x is replaced by multiplication by the Fourier frequency ϕ .

The transformed P_j equation

Transforming term by term gives

$$-\hat{P}_{j,\tau} + ir\phi\hat{P}_j + a\hat{P}_{j,v} + v \left(\alpha_j\hat{P}_j - \beta_j\hat{P}_{j,v} + \gamma\hat{P}_{j,vv} \right) = 0,$$

where

$$\gamma = \frac{\zeta^2}{2}, \quad \alpha_j = iu_j\phi - \frac{\phi^2}{2}, \quad \beta_j = b_j - \rho\zeta i\phi.$$

Because the initial condition is a Heaviside step function, its Fourier representation is understood in the generalized sense. For $\phi \neq 0$, the non-constant part is

$$\hat{P}_j(\phi, v, 0) = \frac{e^{-i\phi \ln K}}{i\phi}.$$

The zero-frequency contribution is recovered by the $\frac{1}{2}$ term in the final Fourier-inversion formula.

Step 6: the exponential-affine ansatz

The transformed initial condition is independent of v , while the transformed PDE is affine in v .

Use

$$\hat{P}_j(\phi, v, \tau) = \exp\{C_j(\tau, \phi)\bar{v}^* + D_j(\tau, \phi)v\} \hat{P}_j(\phi, v, 0).$$

Substitution gives one term independent of v and one term proportional to v . Since the equation must hold for every v , both must vanish separately:

$$\frac{\partial C_j}{\partial \tau} = \frac{ir\phi}{\bar{v}^*} + \kappa^* D_j, \quad \frac{\partial D_j}{\partial \tau} = \gamma D_j^2 - \beta_j D_j + \alpha_j,$$

with $C_j(0, \phi) = D_j(0, \phi) = 0$.

The Riccati equation

The equation for D_j is a Riccati equation:

$$\frac{\partial D_j}{\partial \tau} = \gamma D_j^2 - \beta_j D_j + \alpha_j.$$

Define

$$d_j = \sqrt{\beta_j^2 - 4\gamma\alpha_j}, \quad r_{j,\pm} = \frac{\beta_j \pm d_j}{\zeta^2},$$

with the square-root branch chosen so that $\Re(d_j) \geq 0$.

Then

$$\gamma D^2 - \beta_j D + \alpha_j = \gamma(D - r_{j,+})(D - r_{j,-}).$$

This factorisation makes the nonlinear ODE separable. Once D_j is known, the equation for C_j is linear.

Step 7: closed-form C_j and D_j

Let

$$g_j := \frac{r_{j,-}}{r_{j,+}} = \frac{\beta_j - d_j}{\beta_j + d_j}.$$

Then

$$D_j(\tau, \phi) = r_{j,-} \frac{1 - e^{-d_j \tau}}{1 - g_j e^{-d_j \tau}},$$

and

$$C_j(\tau, \phi) = \frac{ir\phi}{\bar{v}^*} \tau + \kappa^* \left[r_{j,-} \tau - \frac{2}{\zeta^2} \ln \left(\frac{1 - g_j e^{-d_j \tau}}{1 - g_j} \right) \right].$$

These are the closed-form coefficients used in the numerical implementation.

Step 8: recover P_1 , P_2 and the option price

Fourier inversion gives

$$P_j(x, v, \tau) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{\exp(C_j(\tau, \phi) \bar{v}^* + D_j(\tau, \phi) v + i\phi x - i\phi \ln K)}{i\phi} \right] d\phi.$$

Finally,

$$C(t, S, v) = SP_1(t, x, v) - Ke^{-r\tau} P_2(t, x, v).$$

The solution is therefore semi-analytic: C_j and D_j are available in closed form, while the final one-dimensional Fourier integrals are evaluated numerically.

Baseline numerical setup

The numerical implementation uses

$$S_0 = 100, \quad K = 100, \quad r = 0.04, \quad q = 0, \quad T = 1,$$

with

$$v_0 = 0.04, \quad \bar{v}^* = 0.04, \quad \kappa^* = 2, \quad \zeta = 0.3, \quad \rho = -0.7.$$

The risk-neutral Feller condition is satisfied:

$$2\kappa^*\bar{v}^* = 0.16 > \zeta^2 = 0.09.$$

The initial volatility is $\sqrt{v_0} = 20\%$.

Numerical evaluation of the Heston price

Using a Fourier truncation bound $U = 100$ gives

$$P_1 \approx 0.674107, \quad P_2 \approx 0.599492.$$

Therefore

$$C(0, S_0, v_0) = S_0 P_1 - K e^{-rT} P_2 \approx 9.812195.$$

As a benchmark, Black–Scholes using the initial volatility $\sigma_0 = \sqrt{v_0} = 20\%$ gives $C_{\text{BS}} \approx 9.925054$.

Checking Fourier-integral convergence

The pricing formula integrates over $[0, \infty)$, so numerically we replace the upper limit by a finite U and verify stability.

U	25	50	75	100	150	200
C_H	9.811873	9.812190	9.812195	9.812195	9.812195	9.812195

The price is stable to six decimal places by approximately $U = 75$. We therefore use $U = 100$ for the remaining calculations.

How correlation changes the volatility smile

With the numerical implementation stable, keep all other parameters fixed and vary only

$$\rho \in \{0, -0.3, -0.7\}.$$

For each value of ρ :

1. compute Heston call prices across strikes,
2. invert each price through Black–Scholes to obtain implied volatility,
3. plot implied volatility against log-forward moneyness.

The horizontal axis is

$$\log \left(\frac{K}{F_0} \right), \quad F_0 = S_0 e^{(r-q)T}.$$

Thus zero corresponds to the at-the-money-forward point.

Implied-volatility curves

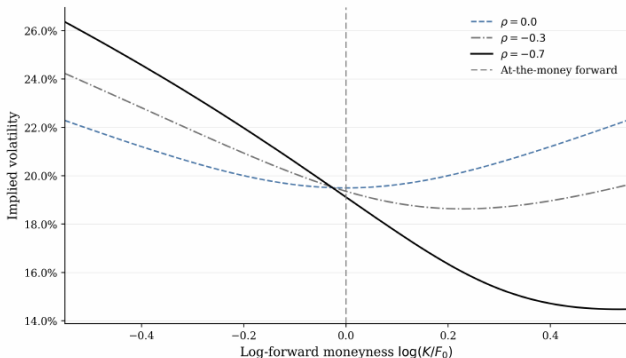


Figure 1: Heston implied-volatility curves as stock–variance correlation changes.

From a symmetric smile to equity skew

When $\rho = 0$, the implied-volatility curve is approximately symmetric around the at-the-money-forward point.

As ρ becomes more negative,

- ▶ implied volatility rises for lower strikes,
- ▶ implied volatility falls for higher strikes,
- ▶ the smile progressively tilts into a downward skew, or smirk.

The effect is strongest for $\rho = -0.7$. In this experiment, stochastic variance generates curvature, while negative stock–variance correlation generates the asymmetry.

Why negative correlation matters

When $\rho < 0$, negative stock-return shocks tend to be associated with positive variance shocks.

This means downside states tend to coincide with higher volatility.

Lower-strike options therefore carry higher implied volatility than higher-strike options, producing the downward skew commonly observed in equity-option markets.

The Heston story

The model starts from a simple economic observation: volatility is not constant.

Allowing variance to evolve stochastically produces a richer implied-volatility structure. Allowing stock and variance shocks to be correlated lets that structure become asymmetric.

At the same time, the CIR variance process gives the model an affine structure. After the log-price transformation and price decomposition, Fourier transformation reduces the P_j equations to a Riccati ODE for D_j and a linear ODE for C_j .

Solving those ODEs gives the transformed functions $\hat{P}_j$; Fourier inversion then recovers P_1 and P_2 , and therefore the European option price.

Key takeaway

The Heston model combines two features that make it especially useful for option pricing:

- ▶ **Economic flexibility:** stochastic, mean-reverting variance and stock–variance correlation can generate the curvature and asymmetry observed in implied-volatility curves.
- ▶ **Mathematical tractability:** the affine CIR structure turns a two-state pricing PDE into a solvable Riccati/linear ODE system plus one-dimensional Fourier integration.

That is the central reason Heston remains a benchmark stochastic-volatility model.

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